Restoration of Historic Oases

Combatting Climate Change & Advancing the Green Economy
in Egypt's Western Desert

Operational Implementation Strategy

Budget: €535,000 **Timeline:** 24 Months

Siwa · **Fayoum** · **Kharga** · **Dakhla**

2026–2028

The Legacy & The Current Situation

For millennia, the historic oases of Egypt's Western Desert have stood as ecological miracles. Sustained by indigenous wisdom and highly adapted water management systems, these "islands of green" have allowed human civilization and unique agro-biodiversity to thrive in one of the harshest environments on Earth.

Today, they face an unprecedented, existential threat.

During our **Forensic Agrarian Review** conducted in the inception phase, field teams documented a critical reality: a severe **"Phenological Shift."** Climate change is physically altering the seasonal cycles and harvest dates of staple crops like dates and olives, throwing traditional farming into chaos and threatening local food security.

Simultaneously, fossil aquifers are depleting, soil salinity is rising rapidly (particularly in Siwa), and the ancestral *"Knowledge by Sight"*—the intricate, unwritten methods of managing scarce water resources—is dying out with the elder generation.

> If we do not act immediately to bridge ancient heritage with modern science, these green havens will be consumed by the desert.

The Vision: Living Laboratories

We refuse to treat these oases as passive museums of a bygone era. UNESCO's vision is to transform them into active **Agro-Ecological Innovation Hubs**—or "Living Laboratories."

By merging 4,000-year-old resilience techniques (like the Fallow Irrigation systems we documented at Ramad Farm) with 21st-century biotechnology (such as microbial fertilizers and tissue culture), we are not just conserving the past; we are engineering a climate-proof future.

Aligning with National & Global Mandates

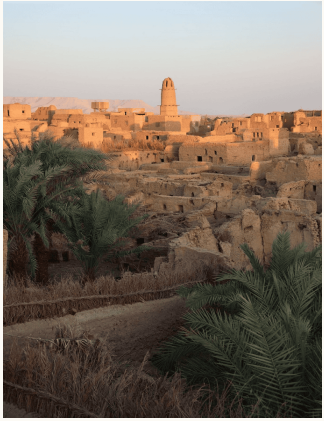
This portfolio is a meticulously designed implementation vehicle for the Egyptian Government's highest strategic commitments:

- **Egypt's NDCs (Climate Action):** Direct operationalization of *"Adaptation in Agriculture"* by promoting drought-resistant crops and securing genetic diversity.
- **Egypt Vision 2030 (Pillar 3):** Rationalizing water use and safeguarding environmental heritage.
- **Climate Strategy 2050 (Goal 2):** Enhancing the climate resilience of vulnerable food systems.
- **UNESCO 43 C/5 Priorities:** Fulfilling our global mandate by integrating the **IHP** (Water Security) and **MAB** (Man & Biosphere) programs on the ground.

Directly Advancing Sustainable Development Goals:

Action on the Ground: The 4 Pilots

We are moving directly from theory to implementation with four shovel-ready interventions, each tailored to the specific climatic and ecological challenges of the target oases.



Dakhla: Water Harvesting

Challenge: The ancient Nubian Aquifer is fading, drying up life-giving channels.
Action: Applying UNESCO's IHP principles, we are physically revitalizing ancient water systems. We will rehabilitate **2-3 traditional wells**, construct **2 sand dams** and a recharge trench to capture runoff, and install **3 hectares** of highly efficient gravity-fed irrigation to sustain agricultural productivity.

Fayoum: Agro-Innovation

Challenge: Severe soil depletion and the loss of heritage farming efficiency.
Action: We are deploying the **Ramad Farm Model** across 10 hectares. This involves reviving ancient "Fallow Irrigation" and intercropping date palms with legumes and medicinal crops (moringa, henna). Crucially, we are introducing university-developed **bio-polymers** to drastically increase soil water retention.

Siwa: Salinity Management

Challenge: Rising, salty groundwater is seeping into fields, poisoning crops and threatening historic sites.

Action: We are implementing bioengineering solutions. We will plant **8 hectares** of highly salt-tolerant heritage crops and establish **150 agro-forestry windbreak trees** to stabilize the soil. Additionally, 2 wetland strips will be developed for natural water filtration and ecosystem recovery.

Kharga: Eco-Tourism & TVET

Challenge: A lack of green economic opportunities drives youth migration to cities.

Action: We are restoring physical heritage to build human capital. We will restore a historic 1960s Administrative Complex into a **School of Traditional Crafts** and convert a mudbrick house into a visitor center. We will train local youth as guides and develop **2 heritage geo-tourism experiences**.

Science & The Circular Economy

To guarantee long-term survival, ecological restoration must be anchored by hard science and economic profitability.

The Genetic Ark: Tissue Culture Propagation
Through our technical partnership with the **Desert Research Center (DRC)**, we identified a dormant Tissue Culture Laboratory during our inception phase. We are providing the operational funds to reactivate this critical asset.

Instead of relying on imported, non-native seeds, this lab will serve as a genetic ark. We will bio-securely mass-propagate **50,000 seedlings** of endangered, climate-resilient heritage date varieties (Saidi and Siwi). This secures the genetic future of the oases against climate shocks.

Breaking the "Grant Trap": A Circular Bio-Economy

Our economic audit revealed that many local NGOs (such as the *Rowad Al-Thaqafa Center*) are trapped in endless grant-dependency cycles. This project shifts them into a self-sustaining **Business Incubation Model**.
We are training and equipping **20 women-led startups** to valorize agricultural waste. By converting discarded date pits and coffee grounds into high-value organic flour, animal feed, and natural cosmetics, we ensure that conservation generates its own revenue long after the project ends.

A Unified Coalition for Resilience

True sustainability requires deep institutional roots. We have moved beyond informal agreements and secured a formally ratified **"5 Pillars of Cooperation" Roadmap** that legally binds our partners. This ensures project continuity from the highest ministries down to the local farmers.

1. National Policy & Executive Authority

Ensuring policy alignment, land access, and cross-sectoral coordination.

- **Ministries:** Irrigation & Water Resources (MWRI), Agriculture (MALR), Environment (MoE), and Tourism & Antiquities (MOTA).
- **Local Executive:** Governorates of New Valley, Fayoum, and Matrouh.

2. Scientific & Technical Engines

Providing the research, data, and biotechnological solutions.

- **Fayoum University:** Developing bio-polymers and agroforestry testing.
- **Desert Research Center (DRC):** Leading tissue culture and salinity mapping.
- **Egyptian Space Agency (EgSA):** Satellite monitoring of vegetation health.
- **New Valley University:** Phenological data tracking.

3. Community & Civil Society

The true stewards of the land, managing daily operations and local knowledge.

- **Water User Associations (WUAs)** in Dakhla.
- **Siwa Community Development Association (CDA)**.
- Local champions: **Ramad Farm**, **Farmers Syndicate**, and the **Rowad Al-Thaqafa Center** (Youth Innovation Hub).

Together, we are engineering a future where heritage saves the environment.

In September 2025, King Felipe VI and President El-Sisi signed the ***Spain–Egypt Alliance for Sustainable Development* (2025–2030).** This project is its direct, operational expression.

Economic Transition
Bilateral trade nearing €3bn · Cairo Metro Line 2 rehabilitation · Egypt–Spain Business Forum

Ecological Transition
Desalination · green hydrogen · sustainable agriculture — *this project is the proof of concept.*

Social Transition
Women's empowerment · institutional modernisation · 13 Spanish archaeological missions in Egypt

Ready for Implementation.

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